

Fractional Linear Transformations

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1 Introduction

In this article, we shall use the following notation. For a given prime p :

- The Finite Field of Integers modulo p is represented by $\mathbb{F}_p$.
- The Multiplicative Group of Integers modulo p is represented by $\mathbb{F}_p^\times$.
- The Projective Space over $\mathbb{F}_p$, denoted by $\mathbb{P}_p$, is the set $\mathbb{F}_p \cup \{\infty\}$.
- The General Linear Group of degree 2 over $\mathbb{F}_p$, denoted by $\text{GL}_2(\mathbb{F}_p)$, is the group of 2×2 invertible matrices with entries over the field $\mathbb{F}_p$.

Definition 1. A Fractional Linear Transformation as a function $f : \mathbb{P}_p \rightarrow \mathbb{P}_p$ of the form

$$f(x) = \frac{ax + b}{cx + d}$$

where $a, b, c, d \in \mathbb{F}_p$.

We abide by the following:

- Take $ad - bc \neq 0$ to avoid f being a constant function.
- Write $f(x) = \infty$ when $cx + d = 0$.
- Write $f(\infty) = \frac{a}{c}$. (which will also be ∞ if $c = 0$)

The last three rules follow intuitively from the conventions of limits over real numbers. For example, when we work over $\mathbb{P}_5$, we have:

$$f(x) = \frac{2x + 1}{x + 3} \rightarrow f(4) = 2$$

We define the set $\text{Möb}(p)$ as the set of all Fractional Linear Transformations for a given prime p . For example, we have:

$$\text{Möb}(2) = \left\{ x, x + 1, \frac{1}{x}, \frac{1}{x + 1}, \frac{x}{x + 1}, \frac{x + 1}{x} \right\}$$

Consider two Fractional Linear Transformations $f(x) = \frac{ax+b}{cx+d}$ and $g(x) = \frac{a'x+b'}{c'x+d'}$. Direct computation gives

$$(f \circ g)(x) = \frac{(aa' + bc')x + (ab' + bd')}{(ca' + dc')x + (cb' + dd')}$$

and

$$f^{-1}(x) = \frac{dx - b}{-cx + a}$$

To check that $f \circ g$ is not a constant function, it suffices that $(aa' + bc')(cb' + dd') - (ab' + bd')(ca' + dc') = (ad - bc)(a'd' - b'c') \neq 0$ since neither f nor g were constant functions. Similarly, $f^{-1}(x)$ is not constant because $da - (-b)(-c) = ad - bc \neq 0$ since f is not constant. This shows that $\text{Möb}(p)$ is closed under composition and inversion. Additionally, consider the function $\iota(x) = x$. Composing with another Fractional Linear Transformation yields $(f \circ \iota)(x) = (\iota \circ f)(x) = f(x)$. Hence, $\text{Möb}(p)$ can be considered a group with function composition as group multiplication and ι as the identity.

The formulas for inversion and composition of Fractional Linear Transformations motivates the following definition:

Definition 2. For a Fractional Linear Transformation f , the associated matrix $F \in \text{GL}_2(\mathbb{F}_p)$ is defined as follows:

$$f(x) = \frac{ax + b}{cx + d} \quad \leftrightarrow \quad F = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

For the remainder of the article, Fractional Linear Transformations will be denoted with lowercase Roman letters and their associated matrices will be denoted by uppercase Roman letters. Composition of a Fractional Linear Transformation with itself will be denoted by exponents e.g. $f^2(x) = (f \circ f)(x)$.

Note that we can also consider kF as an associated matrix of f , for any $k \in \mathbb{F}_p^\times$ since the scaling factor cancels in the numerator and denominator of $f(x)$. The reason behind this correspondence will be made clear later on when discussing properties of the associated matrices (see Eigenvalues and Eigenvectors).

It is easy to verify the following relations with associated matrices:

- Identity: The associated matrix of the identity transformation, ι , is the identity matrix.
- Composition: If f_1 and f_2 are Fractional Linear Transformations with associated matrices F_1 and F_2 respectively, then the associated matrix of $f_1 \circ f_2$ is $F_1 F_2$.
- Inverse: If f is a Fractional Linear Transformation with associated matrix F , then the associated matrix of f^{-1} is F^{-1} .

Every element of $\text{GL}_2(\mathbb{F}_p)$ is the associated matrix of exactly one Fractional Linear Transformation. It follows that there is a surjective map $\varphi : \text{GL}_2(\mathbb{F}_p) \rightarrow \text{Möb}(p)$. Using the three properties above, we can conclude that φ is a surjective homomorphism i.e. $\text{Möb}(p)$ is a normal subgroup of $\text{GL}_2(\mathbb{F}_p)$.

The kernel of the above homomorphism is the set of associated matrices sent to $\iota(x) = x$, and it follows that:

$$\ker(\varphi) = \left\{ \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix} : n \in \mathbb{F}_p^\times \right\} \implies \ker(\varphi) \cong \mathbb{F}_p^\times$$

Since $\text{Möb}(p)$ is a quotient group derived from the above homomorphism, we have:

$$\text{Möb}(p) \cong \text{GL}_2(\mathbb{F}_p) / \ker(\varphi) \implies \text{Möb}(p) \cong \text{GL}_2(\mathbb{F}_p) / \mathbb{F}_p^\times$$

Finally, we can compute the number of Fractional Linear Transformations given prime p . We know that $\text{GL}_2(\mathbb{F}_p)$ has $(p^2 - 1)(p^2 - p) = (p - 1)^2 p(p + 1)$ elements and $\mathbb{F}_p^\times$ has $p - 1$ elements. It follows:

$$|\text{Möb}(p)| = \frac{|\text{GL}_2(\mathbb{F}_p)|}{|\mathbb{F}_p^\times|} = \frac{(p - 1)^2 p(p + 1)}{(p - 1)} = (p - 1)p(p + 1) = p^3 - p$$

2 Cycle Structure

Consider elements $x, y \in \mathbb{P}_p$ such that $f(x) = f(y)$ for some $f \in \text{Möb}(p)$. It then follows that we have:

$$\frac{ax+b}{cx+d} = \frac{ay+b}{cy+d} \implies (ad-bc)(x-y) = 0$$

Since $ad-bc \neq 0$, we must have $x = y$, i.e. the function is injective. Moreover, the domain and the codomain have the same size. Hence, every Fractional Linear Transformation is bijective.

It follows that every function $f \in \text{Möb}(p)$ acts as a permutation of the elements of $\mathbb{P}_p$. It then makes sense to try and classify linear fractional transformations based on common properties of their associated permutations.

The simplest structure to study on the permutation of $\mathbb{P}_p$ is the fixed points of f . The fixed points are determined as solutions to $f(x) = x$. If $f(x) = \frac{ax+b}{cx+d}$, the equation becomes

$$\begin{aligned} \frac{ax+b}{cx+d} &= x \\ cx^2 + (d-a)x + b &= 0 \end{aligned}$$

and the fixed points of $f(x)$ amount to solving a quadratic equation. For the identity function, all coefficients of the quadratic are 0 so any x is a solution. By Lagrange's theorem on polynomial equations, for all non-identity functions, if all the fixed points are in $\mathbb{F}_p$, there are at most 2 roots. If ∞ is a fixed point, since $f(\infty) = \frac{a}{c}$ in general, c must be 0. Hence, the quadratic equation for the rest of the roots in $\mathbb{F}_p$ reduces to a linear equation which has at most one additional root. For the identity transformation alone, we have more than 2 fixed points (all are fixed). Thus, we have the following:

Theorem 2.1. *The orbit diagram of any $f \in \text{Möb}(p)$ has at most two fixed points, with the exception of the identity function, ι .*

We now proceed to study the other components of the orbit diagram of Fractional Linear Functions.

Definition 3. *A n -cycle of $f(x)$ is a sequence $a_1, \dots, a_n$ of distinct elements of $\mathbb{P}_p$ such $f(a_i) = a_{i+1}$ for $1 \leq i < n$ and $f(a_n) = a_1$.*

If f has an n -cycle $a_1, \dots, a_n$, it is easy to verify that $f^n(a_i) = a_i$ for all elements of the n -cycle. If $n > 2$, $f^n(x)$ has more than 2 fixed points and hence $f^n = \iota$. Assume that $f(x)$ has an n -cycle $a_1, \dots, a_n$ and an m -cycle $b_1, \dots, b_m$ for $2 < m \leq n$. It follows that since $f^n = \iota$, $f^n(b_i) = b_i$ but this is only possible if $m|n$. By symmetry, $n|m$ so $m = n$. This implies that if f has an n -cycle for $n > 2$, it is composed of 2-cycles, n -cycles (for a fixed $n > 2$) and at most two fixed points.

Assume now that f has a 2-cycle a_1, a_2 and an n -cycle for $n > 2$. If f had a fixed point b , then f^2 would have three fixed points, namely a_1, a_2 , and b . This would imply $f^2 = \iota$ but then f cannot have an n -cycle. Hence, $f(x) = x$ must have no solutions in $\mathbb{P}_p$ but $f^2(x) = x$ must have at least two solutions in $\mathbb{P}_p$, namely a_1, a_2 . This implies at least one of a_1, a_2 is in $\mathbb{F}_p$ and must be a solution to a certain quadratic equation. Let $f(x) = \frac{ax+b}{cx+d}$. Then, $f(x) = x$ reduces to $cx^2 + (d-a)x + b = 0$ which has no solutions in $\mathbb{P}_p$ only if $c \neq 0$ and the discriminant is a quadratic non-residue in $\mathbb{F}_p$. The discriminant of $f(x) = x$ is $(d-a)^2 - 4bc$. Now, consider $f^2(x)$:

$$f^2(x) = (f \circ f)(x) = f\left(\frac{ax+b}{cx+d}\right) = \frac{(a^2+bc)x + b(a+d)}{c(a+d)x + (d^2+bc)}$$

The discriminant of the fixed point equation $f^2(x) = x$ is

$$(d^2 - a^2)^2 - 4bc(a+d)^2 = (a+d)^2((d-a)^2 - 4bc)$$

which must be a quadratic residue mod p to ensure solutions. However, the discriminant of $f^2(x) = x$ is the product of a quadratic residue and the discriminant of $f(x) = x$. Since the product of a quadratic

residue with a quadratic non-residue is a quadratic non-residue, we conclude that either both $f(x) = x$ and $f^2(x) = x$ have solutions over $\mathbb{F}_p$ or neither have solutions over $\mathbb{F}_p$. This means that $f(x)$ cannot have a 2-cycle and an n -cycle for $n > 2$. From these facts, we obtain the following theorem

Theorem 2.2. *A non-identity Fractional Linear Transformation $f(x) = x$ forms a permutation on $\mathbb{P}_p$ that decomposes into a combination of n -cycles for some fixed $n \geq 2$ and at most two fixed points. This value of n will be called the order of $f(x)$.*

If a Fractional Linear Transformation $f(x)$ has no fixed points and splits $\mathbb{P}_p$ into n -cycles, since $\mathbb{P}_p$ has $p + 1$ elements, $n|p + 1$. Similarly, if $f(x)$ has exactly one fixed point, it splits the rest of $\mathbb{P}_p$ into n -cycles where $n|p$, but since p is prime, $n = p$. Lastly, if $f(x)$ has two fixed points, it splits the rest of $\mathbb{P}_p$ into n -cycles where $n|p - 1$.

Hence, any Fractional Linear Transformation can be put into one of four families:

- The identity function
- No fixed points and n -cycles where $n|p + 1$
- One fixed point and p -cycles
- Two fixed points and n -cycles where $n|p - 1$

The next sections will illuminate further the properties of functions within each of these families in relation to one another.

3 Eigenvalues and Eigenvectors

The space $\mathbb{P}_p$ can be identified with $(\mathbb{F}_p^2 - 0)/\mathbb{F}_p^\times$. The element $a \in \mathbb{F}_p$ is associated with $[a, 1]^T$ and the element ∞ is associated $[1, 0]^T$ where $[km, kn]^T$ represents the same element as $[m, n]^T$ for all $k \in \mathbb{F}_p$. It should be noted that $(\mathbb{F}_p^2 - 0)/\mathbb{F}_p^\times$ is not a vector space since it does not contain an additive identity, and that addition of these vectors does not in general correspond to addition of the corresponding elements of $\mathbb{P}_p$.

Let F be the associated matrix of $f(x) = \frac{ax+b}{cx+d}$. Then,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} ax + b \\ cx + d \end{pmatrix} = \begin{pmatrix} \frac{ax+b}{cx+d} \\ 1 \end{pmatrix}$$

and

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} \frac{a}{c} \\ 1 \end{pmatrix}$$

where the resultant vector is considered $[1, 0]^T$ if $cx + d = 0$ or $c = 0$ respectively. Hence, the action of F on $(\mathbb{F}_p^2 - 0)/\mathbb{F}_p^\times$ is identical to the action of $f(x)$ on $\mathbb{P}_p$.

An eigenvector of a matrix M over a field K is a non-zero vector v such that $Mv = \lambda v$ for some $\lambda \in K$ and λ is called the associated eigenvalue. The eigenvalues of any invertible matrix are non-zero. If $v \in (\mathbb{F}_p^2 - 0)/\mathbb{F}_p^\times$ is an eigenvector of $F \in \text{GL}_2(\mathbb{F}_p)$, the corresponding point $a \in \mathbb{P}_p$ is a fixed point of the corresponding transformation $f(x)$ since λv corresponds to the same point in $\mathbb{P}_p$ as v .

The eigenvalues are zeros of the characteristic polynomial of F , $\det(F - \lambda \mathbb{I})$, where $\mathbb{I}$ is the identity matrix. For 2×2 matrices, the equation becomes $\lambda^2 - \lambda \text{tr}(F) + \det(F) = 0$. For $p > 2$, using the quadratic formula, the eigenvalues of F are $\lambda = \frac{\text{tr}(F) \pm \sqrt{(\text{tr}(F))^2 - 4 \det(F)}}{2}$. Note that the discriminant is the same as the discriminant of $f(x) = x$.

If the discriminant, $(\text{tr}(F))^2 - 4 \det(F)$, is a non-zero quadratic residue over $\mathbb{F}_p$, there are two distinct eigenvalues of F in $\mathbb{F}_p$. It follows that there are two linearly independent eigenvectors of F and hence, $f(x)$ has two fixed points.

If the discriminant is 0, there is exactly one eigenvalue of F . There are two cases: F has two linearly independent eigenvectors or F has one eigenvector. If F has two linearly independent eigenvectors v_1, v_2 , since v_1 and v_2 span the entire vector space $\mathbb{F}_p^2$, it follows that $F = \lambda \mathbb{I}$. However, if F is a multiple of the identity, the corresponding transformation must be the identity transformation. If F has one eigenvector, $f(x)$ has only one fixed point.

If the discriminant is a quadratic non-residue, F has no eigenvalues over $\mathbb{F}_p$. In this case, it is useful to consider a field extension of $\mathbb{F}_p$ over which F has eigenvalues. If $p > 2$, there are $\frac{p-1}{2}$ quadratic residues and $\frac{p-1}{2}$ quadratic non-residues. Since multiplying a quadratic residue by a quadratic non-residue is a quadratic non-residue in $\mathbb{F}_p$, multiplication by a fixed quadratic non-residue k forms a bijection between quadratic residues and quadratic non-residues. The field $\mathbb{F}_p(\sqrt{k})$ then has a square root for all elements in $\mathbb{F}_p$. This means that if a matrix $F \in \text{GL}_2(\mathbb{F}_p)$ has no eigenvalues in $\mathbb{F}_p$, it has conjugate eigenvalues $a \pm \sqrt{k}$ in $\mathbb{F}_p(\sqrt{k})$. Note that this is no different from taking a number $i = \sqrt{-1}$ to ensure that every element of $\mathbb{R}$ has a square root in $\mathbb{R}(i) = \mathbb{C}$.

Theorem 3.1. *If $p > 2$ and k is a quadratic non-residue in $\mathbb{F}_p$, $\mathbb{F}_p(\sqrt{k}) = a + b\sqrt{k}$ for $a, b \in \mathbb{F}_p$.*

Proof. It is easy to see that $a + b\sqrt{k}$ is closed under addition and multiplication and that every element has an additive inverse. All $a \neq 0$ have a multiplicative inverse from $\mathbb{F}_p$. For an element $a + b\sqrt{k}$, $b \neq 0$, consider $(a + b\sqrt{k})(a - b\sqrt{k}) = a^2 - kb^2$. The last quantity is an element of $\mathbb{F}_p$ and is not zero for if it were, $k = (ab^{-1})^2$ would be a quadratic residue mod p , contrary to assumption. Hence, the multiplicative inverse of $a + b\sqrt{k}$ is $\frac{a-b\sqrt{k}}{a^2-kb^2}$. $\square$

The case $p = 2$ is more subtle because there are no quadratic non-residues and using the quadratic formula directly involves dividing by zero. The characteristic equations of matrices in $\text{GL}_2(\mathbb{F}_2)$ are either linear or monic quadratic. By enumerating all such equations, it is easy to see that the only one with no solutions in $\mathbb{F}_2$ is $x^2 + x + 1 = 0$. A suitable field extension is then $\mathbb{F}_2(\omega)$ for $\omega^2 + \omega + 1 = 0$. It is easy to show $\mathbb{F}_2(\omega) = a + b\omega + c\omega^2$ for $a, b, c \in \mathbb{F}_2$.

Now, any matrix $F \in \text{GL}_2(\mathbb{F}_p)$ has either one eigenvalue or two eigenvalues in some field extension K of $\mathbb{F}_p$. If F has one eigenvalue, f is either the identity function or has exactly one fixed point. If f has one fixed point, it must permute the other p elements of $\mathbb{F}_p$. Since p is prime, the structure of f is that of one fixed point and one p -cycle.

All other $F \in \text{GL}_2(\mathbb{F}_p)$ have two eigenvalues $\lambda_1, \lambda_2 \in K$ and hence two non-zero linearly independent eigenvectors v_1, v_2 in K^2 . Let P be the matrix with columns v_1, v_2 and D be the diagonal matrix $\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$. F can then be diagonalized as $F = PDP^{-1}$. Let n be the smallest integer so that $F^n = m\mathbb{I}$ for some $m \in \mathbb{F}_p^\times$. We know that $2 \leq n \leq p+1$ because it is the order of the corresponding function f . Then,

$$\begin{aligned} F^n &= (PDP^{-1})^n = PD^nP^{-1} = m\mathbb{I} \\ D^n &= P^{-1}(m\mathbb{I})P = m\mathbb{I} \\ \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} &= \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \end{aligned}$$

It then follows that $(\lambda_1\lambda_2^{-1})^n = mm^{-1} = 1$. Hence, the order of f in $\text{Möb}(p)$ is the order of the ratio of the corresponding eigenvalues in K .

Consider the example $f(x) = \frac{4x+2}{x+4}$. The associated matrix is $F = \begin{pmatrix} 4 & 2 \\ 1 & 4 \end{pmatrix}$. The characteristic equation of F in $\mathbb{P}_5$ is $\lambda^2 - 3\lambda + 4 = 0$ which has solutions $\lambda = \frac{3 \pm \sqrt{3}}{2}$. Since 3 is a quadratic non-residue mod 5, we conclude that F has no eigenvalues in $\mathbb{F}_5$ and hence $f(x)$ has no fixed points. Now, consider the field extension $K = \mathbb{F}_5(\sqrt{2})$. The eigenvalues can then be written as

$$\lambda = \frac{3 \pm \sqrt{3}}{2} = \frac{3 \pm \sqrt{4 \times 2}}{2} = 4 \pm \sqrt{2}$$

which are conjugate eigenvalues in K . Now, the ratio of the eigenvalues is $2 + \sqrt{2}$ which has order 3 in K . We conclude that f is composed of two 3-cycles (since there are 6 elements in $\mathbb{P}_5$). It can be checked that f is composed of the cycles $\{0, 3, 2\}$ and $\{\infty, 4, 1\}$ as expected.

4 Families and Conjugacy Classes

4.1 Conjugation and Permutations

Definition 4. The conjugacy classes of a group G is the partition of the set of group elements using the equivalence relation:

$$g \sim h \iff \exists a \in G; g = aha^{-1}$$

We first analyze the effect of conjugation in the symmetric group. Consider permutations $\tau, \sigma \in S_n$.

Theorem 4.1. The permutation $\sigma\tau\sigma^{-1}$ has the same cycle decomposition as τ , with the elements in the cycle decomposition permuted by σ .

Proof. Since σ is a bijection, the statement is equivalent to proving that if τ sends x to y , then $\sigma\tau\sigma^{-1}$ sends $\sigma(x)$ to $\sigma(y)$. We have:

$$\tau(x) = y \implies (\sigma\tau\sigma^{-1})(\sigma(x)) = \sigma(\tau(x)) = \sigma(y)$$

which proves the required. □

Corollary 4.1.1. The structure of cycle decomposition is invariant under conjugation.

Corollary 4.1.2. The element x is a fixed point of τ if and only if $\sigma(x)$ is a fixed point of $\sigma\tau\sigma^{-1}$.

Corollary 4.1.3. The number of fixed points is invariant under conjugation.

We recall the subdivision of $\text{Möb}(p)$ into 4 families based on the number of fixed points. It follows from the last corollary that these families are disjoint unions of conjugacy classes. Hence, we may study the structure of each of these families and their conjugacy classes.

A really important manipulation that we shall implement repeatedly is the conjugation of Fractional Linear Transformations. We have four coefficients a, b, c, d , and the effective choice of three coefficients after scaling. It follows that for any choice $x, x', y, y' \in \mathbb{P}_p$, we can find $f \in \text{Möb}(p)$ such that $f(x) = x'$ and $f(y) = y'$ since we have an equation with three unknowns and only two input-output pairs. Since every non-identity transformation has at most two fixed points, we may now use conjugation to alter the fixed points to our choice and yield a new fractional linear transformation.

Now, to study the conjugacy classes of $\text{Möb}(p)$, all we have to do is study representatives from each conjugacy class with fixed points chosen to our convenience. This proves to be helpful as specific choices for fixed points helps in getting a better understanding of the conjugacy classes themselves through representatives. The method is also very helpful in counting arguments.

Before going into the different families, we outline the use of this conjugation property in the following theorem.

Theorem 4.2. The number of functions $f \in \text{Möb}(p)$ that have the element m as a fixed point is $p(p-1)$.

Proof. Here, we present two independent proofs, one using the special properties of Fractional Linear Transformations that have been discussed and another using more general properties from group theory.

First, consider conjugating f by $g(x) = \frac{mx+1}{x}$ which sends ∞ to m . It is easy to see $g \circ f \circ g^{-1}$ sends ∞ to ∞ so without loss of generality, we may assume that $m = \infty$. If ∞ is a fixed point of f , $c = 0$ and we may scale to make $f(x) = ax + b$. Now, we may have a be any non-zero element of $\mathbb{F}_p$ and b be any element of $\mathbb{F}_p$. This leads to a total of $p(p-1)$ elements that fix ∞ . Hence there are $p(p-1)$ elements that fix m .

The second proof relies on the orbit-stabilizer theorem. Let $\text{Stab}(m) = \{f \in \text{Möb}(p) \mid f(m) = m\}$ and $\text{Orb}(m) = \{x \in \mathbb{F}_p \mid \exists f \in \text{Möb}(p), f(m) = x\}$. The orbit stabilizer theorem tells us $|\text{Stab}(m)| \cdot |\text{Orb}(m)| = |\text{Möb}(p)|$. We have shown in the previous proof that ∞ can be sent to any point and hence $\infty \in \text{Orb}(m)$ for all $m \in \mathbb{F}_p$. It is also easy to see that $n \in \text{Orb}(m)$ for all $n \in \mathbb{F}_p$ by taking $f(x) = x + (n - m)$ which is non-singular. It follows that $|\text{Orb}(m)| = |\mathbb{F}_p| = p + 1$ and $|\text{Stab}(m)| = p(p - 1)$ as expected. $\square$

4.2 Family with 2 fixed points

Theorem 4.3. *The size of the family with 2 fixed points is $\frac{(p-2)p(p+1)}{2}$.*

Proof. We first use conjugation to make our fixed points as 0 and ∞ . Let $f(x)$ such a function. Then,

$$f(x) = \frac{ax + b}{cx + d} \implies \frac{b}{d} = 0; \frac{a}{c} = \infty \implies b = c = 0$$

Now, both a and d are non-zero. As scaling is ignored, we may take $d = 1$, which gives us $f(x) = ax$. Notice that $a \neq 1$ since f is not the identity function. This means that we have $p - 2$ choices for a , and hence, f . Finally, we remove the constraint by allowing any two fixed points. Since we have $p + 1$ elements in $\mathbb{P}_p$ and we are to choose 2 fixed points, it follows that we have:

$$\text{Number of FLTs with 2 fixed points} = (p - 2) \cdot \binom{p + 1}{2} = \frac{(p - 2)p(p + 1)}{2}$$

as required. $\square$

Theorem 4.4. *The number of conjugacy classes in the family of 2 fixed points is $\frac{p-1}{2}$.*

Proof. We know that we can take the representatives from the conjugacy classes of the form $f(x) = ax$. We know that the associated matrix is as follows:

$$f(x) = ax \leftrightarrow F = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

The eigenvalues are a and 1. Hence, $f(x) = ax$ and $f(x) = bx$ can be considered to have the same ratio of eigenvalues if and only if either $a = b$ or $ab = 1$. Moreover, they are in fact conjugate to each other as we have:

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{-1}$$

Hence, we know that $f(x) = ax$ and $f(x) = a^{-1}x$ are the representatives for the conjugacy classes in the family of 2 fixed points. Note that $a \notin \{0, 1\}$ since a is to be non-zero and $f \neq \text{id}$. This means that we have $p - 2$ choices for the value of a . In these choices, $p - 3$ of them form pairs with each pair containing a and a^{-1} . For $a = p - 1$, we have only one representative since it is its own inverse. This is in fact the function $f(x) = -x$ whose orbit diagram is composed of 2-cycles and 2 fixed points.

We have $p + 1$ elements in $\mathbb{P}_p$, and it follows that we can choose $\binom{p+1}{2}$ pairs of fixed points. Multiplying this out by the number of representatives, we have $\frac{p-3}{2}$ conjugacy classes with $2 \cdot \binom{p+1}{2} = p(p + 1)$ elements, and 1 conjugacy class with $\binom{p+1}{2} = \frac{p(p+1)}{2}$ elements. Moreover:

$$\text{Number of conjugacy classes in family of 2 fixed points} = \frac{p-3}{2} + 1 = \frac{p-1}{2}$$

which proves the theorem. $\square$

4.3 Family with 1 fixed point

Theorem 4.5. *The size of the family with 1 fixed point is $(p-1)(p+1)$.*

Proof. We use conjugation to make our fixed point as ∞ . This means that we may take $c = 0$, and due to scaling, we may take $d = 1$. This gives us $f(x) = ax + b$. This cannot have any fixed points other than ∞ , so we must have the fixed point $\frac{-b}{a-1} = \infty \implies a = 1$. Thus, $f(x) = x + b$, where $b \neq 0$.

Since we have $p-1$ choices for b , and have $p+1$ choices for the fixed point after removing the fixed point constraint, it follows:

$$\text{Number of FLTs with 1 fixed point} = (p-1)(p+1)$$

which proves the required. $\square$

Theorem 4.6. *There is exactly 1 conjugacy class in the family with 1 fixed point*

Proof. We simply show that each representative can be expressed by conjugating the representative with $b = 1$:

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} b & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b & 1 \\ 0 & 1 \end{pmatrix}^{-1}$$

It would follow that every pair of representatives, and hence every pair of functions in the family of 1 fixed point are conjugate to one another. Hence, the family as a whole forms 1 conjugacy class. $\square$

4.4 Family of 0 fixed points

Theorem 4.7. *The size of the family with 0 fixed points is $\frac{p^2(p-1)}{2}$.*

Proof. We use the field extension $K = F_p(\sqrt{k})$. For convenience, we use conjugation and take the fixed points as $\pm\sqrt{k}$. We know that the solution of the fixed point equation is:

$$f(x) = \frac{ax+b}{cx+d} = x \implies cx^2 + (d-a)x - b = 0 \implies x = \frac{(a-d) \pm \sqrt{(a-d)^2 + 4bc}}{2c}$$

Due to scaling, we may take $c = 1$ (here, $c \neq 0$ as ∞ is not a fixed point). Then, we have $a-d = 0$ and $b = k$. Thus, our representatives are of the form $f(x) = \frac{ax+k}{x+a}$. We have p choices for a as there are no restrictions on what value it takes.

Since the fixed points of the extended field are solutions of a quadratic equation, it follows that they form a conjugate pair $x \pm y\sqrt{k}$. We have p choices for x and $p-1$ for y since it is non-zero, giving us $\frac{p(p-1)}{2}$ pairs of fixed points. Hence, we have:

$$\text{Number of FLTs with 2 fixed points} = p \cdot \frac{p(p-1)}{2} = \frac{p^2(p-1)}{2}$$

as required. $\square$

Theorem 4.8. *The number of conjugacy classes in the family of 0 fixed points is $\frac{p+1}{2}$.*

Proof. Consider the representatives $f(x) = \frac{ax+k}{x+a}$. The eigenvalues of the associated matrix are $a \pm \sqrt{k}$. If we have the ratio of eigenvalues to be equal for distinct choices a and a' , we would have:

$$\frac{a + \sqrt{k}}{a - \sqrt{k}} = \frac{a' - \sqrt{k}}{a' + \sqrt{k}} \implies a + a' = -(a + a') \implies a' = -a$$

We can check that these representatives in fact belong to the same conjugacy class since we have:

$$\begin{pmatrix} -a & k \\ 1 & -a \end{pmatrix} = - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & k \\ 1 & a \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^{-1}$$

Hence, there are 2 representatives in all but one conjugacy class, with the exception being $a = 0 = -a$. Here, the function $f(x) = \frac{k}{x}$ is composed of 2-cycles. Hence, we have $\frac{p-1}{2}$ conjugacy classes with $2 \cdot \binom{p}{2} = p(p-1)$ elements, and 1 conjugacy class with $\binom{p}{2} = \frac{p(p-1)}{2}$ elements. Moreover:

$$\text{Number of conjugacy classes in family of 0 fixed points} = \frac{p-1}{2} + 1 = \frac{p+1}{2}$$

which proves the theorem. $\square$

4.5 The Structure of the Mobius Group

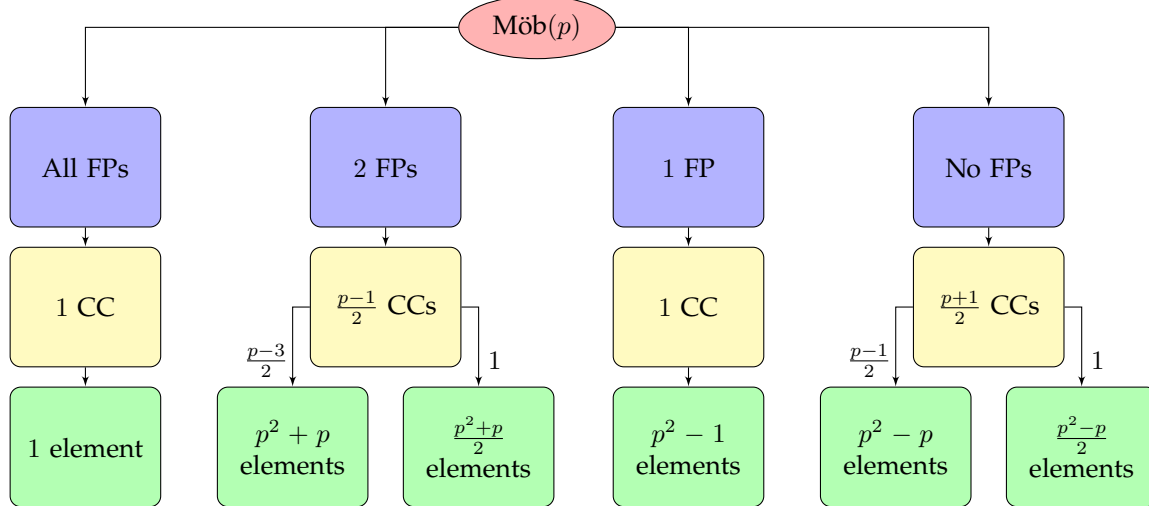
Theorem 4.9. *The number of conjugacy classes in $\text{Möb}(p)$ is $p+2$.*

Proof. Simply adding up, the number of conjugacy classes is:

$$1 + \frac{p-1}{2} + 1 + \frac{p+1}{2} = p+2$$

which is the statement of the theorem. $\square$

We give a visual representation for the structure of $\text{Möb}(p)$ below:



In the each section up to now, we have considered p as an odd prime. The case $p = 2$ is slightly different from the other primes since we have no quadratic non-residues for the field extension, but instead have a field extension formed by the cube root of unity.

For $p = 2$, if a fractional linear transformation has 2 fixed points, since $\mathbb{P}_2$ has 3 elements, it would follow that the transformation is the identity. This means that the family with exactly 2 fixed points is empty (which can be directly seen by the formula for the size of this conjugacy class).

For the family with 1 fixed point, the formula for odd primes again applies and we would have 3 elements in the family, together forming 1 conjugacy class. However, the scenario with the family with 0 fixed points is different due to the choice of field extension being different.

We have two elements in $\mathbb{F}_2(\omega) - \mathbb{F}_2$, the non-trivial cube roots of unity. They are the fixed points to 2 fractional linear transformations. Hence, $\text{Möb}(2)$ has three families, each one being a whole conjugacy

class:

$$\begin{aligned}
\text{Family of 3 fixed points} &\rightarrow \iota(x) = x \\
\text{Family of 1 fixed point} &\rightarrow f_1(x) = \frac{1}{x} \quad ; \quad f_2(x) = x + 1 \quad ; \quad f_3(x) = \frac{x}{x+1} \\
\text{Family of 0 fixed points} &\rightarrow g_1(x) = \frac{x+1}{x} \quad ; \quad g_2(x) = \frac{1}{x+1}
\end{aligned}$$

We can see that since the orbit diagrams of $\text{Möb}(2)$ are the permutations of 3 elements of the projective space and $|\text{Möb}(2)| = 6$, we have $\text{Möb}(2) \cong S_3$.

4.6 The Number of Elements of a Given Order

We know that the number of elements of order 1 is 1 and the number of elements of order p is $(p-1)(p+1)$. These cases are direct from the fact that their respective families have elements of the same order. However, finding the number of elements of order n in the family of 2 fixed points (here, $n \mid p-1$) and the family of 0 fixed points (here, $n \mid p+1$) is much more subtle.

Theorem 4.10. *Let φ denote the Euler Totient Function. Then:*

$$\sum_{d \mid n} \varphi(d) = n$$

Proof. This simply follows from writing the fractions $\frac{i}{n}$ in simplest form for $1 \leq i \leq n$. There are clearly n of these fractions. One other way to count them is by considering how many fractions are there with denominator d . Each such d must be a divisor of n and the numerator would have $\varphi(d)$ choices. This proves the given claim. $\square$

Theorem 4.11. *Two elements in the same conjugacy class have the same order.*

Proof. We take the functions f and $g = afa^{-1}$:

$$f^n = \iota \iff af^n a^{-1} = (afa^{-1})^n = \iota \iff g^n = \iota$$

which proves that f and g have the same order. $\square$

This allows us to instead conveniently study the order of the representatives of the conjugacy classes alone. To find the number of elements, we may later multiply the count by the respective sizes of the conjugacy classes.

Theorem 4.12. *In the family of 2 fixed points, there are $\frac{p(p+1)\varphi(n)}{2}$ elements of order $n > 1$.*

Proof. The representatives are of the form $f(x) = ax$. For the moment, we include ι inside the family. The order of FLT is simply the ratio of the eigenvalues, which here, is a . Now, we know that all of the ratios can be considered as distinct elements of $\mathbb{F}_p$, and the equation $\lambda^{p-1} - 1$ has $p-1$ distinct roots since all elements have order dividing $p-1$ (this is Fermat's Little Theorem). Since $\lambda^n - 1$ is a factor of $\lambda^{p-1} - 1$, it follows that the former must have n distinct roots (this follows from Lagrange's theorem). Hence, there are n representatives with order dividing n .

We now claim that the number of representatives of order n is $\varphi(n)$. This can be done by inducting on the number of prime factors of n . The case $n = 1, 2$ are trivial. To find the number of representatives of order n , we simply add the number of representatives of order d for each $d \mid n$. We know by induction hypothesis that the number of representatives for each $d < n$ is $\varphi(d)$. Hence:

$$\text{Number of representatives of order } n = n - \sum_{d \mid n, d < n} \varphi(d) = n - (n - \varphi(n)) = \varphi(n)$$

Finally, each conjugacy class for $n > 2$ has two representatives, and hence, there are $\frac{1}{2}\varphi(n)$ conjugacy classes with elements of order n . For $n = 2$, we have exactly 1 conjugacy class corresponding to the representative with $a = p - 1$. Next, each representative under conjugation gives a set of $\binom{p+1}{2}$ elements of the family including itself. Hence, for $n > 1$ and $n \mid p - 1$:

$$\text{Number of elements of order } n = \varphi(n) \cdot \binom{p+1}{2} = \frac{p(p+1)\varphi(n)}{2}$$

□

Theorem 4.13. *In the family of 0 fixed points, there are $\frac{p(p-1)\varphi(n)}{2}$ elements of order $n > 1$.*

Proof. This works along the same lines as the previous proof. We include the identity for the moment and we would have the other p ratios of the form $\frac{a+\sqrt{k}}{a-\sqrt{k}}$. All of these are $p+1$ distinct roots of $\lambda^{p+1} - 1$, and the rest of the argument holds in the exact same manner.

There are $\frac{1}{2}\varphi(n)$ conjugacy classes with elements of order n . For $n = 2$, we have exactly 1 conjugacy class corresponding to the representative with $a = 0$. Each representative under conjugation gives a set of $\binom{p}{2}$ elements of the family including itself. Hence, for $n > 1$ and $n \mid p - 1$:

$$\text{Number of elements of order } n = \varphi(n) \cdot \binom{p}{2} = \frac{p(p-1)\varphi(n)}{2}$$

Corollary 4.13.1. *Given a positive integer n , we have:*

$$\text{Number of elements of Möb}(p) \text{ of order } n = \begin{cases} 1 & \text{if } n = 1 \\ p^2 & \text{if } n = 2 \\ \frac{p(p+1)\varphi(n)}{2} & \text{if } n > 2 \text{ and } n \mid (p-1) \\ \frac{p(p-1)\varphi(n)}{2} & \text{if } n > 2 \text{ and } n \mid (p+1) \\ 0 & \text{otherwise} \end{cases}$$

□

5 Recurrence Equations

By the Cayley-Hamilton Theorem, a matrix F satisfies its own characteristic polynomial so that $F^2 - \text{tr}(F)F + \det(F)\mathbb{I} = 0$. Rearranging, $F^2 = \text{tr}(F)F - \det(F)\mathbb{I}$. If $F^n = a_n F + b_n \mathbb{I}$,

$$\begin{aligned} F^{n+1} &= a_n F^2 + b_n F \\ &= a_n (\text{tr}(F)F - \det(F)\mathbb{I}) + b_n F \\ &= a_{n+1} F + b_{n+1} \mathbb{I} \end{aligned}$$

and F^n is a linear function of F . The coefficients b_n satisfy $b_n = -\det(F)a_{n-1}$ and hence, a_n satisfies $a_n = \text{tr}(F)a_{n-1} + b_{n-1} = \text{tr}(F)a_{n-1} - \det(F)a_{n-2}$. Convenient initial values for a_n are $a_0 = 0$ (since $F^0 = \mathbb{I}$) and $a_1 = 1$ (since $F^1 = F$). The coefficients of b_n in fact satisfy the same recurrence relation but with different initial values.

If n is the order of f , $F^n = b_n \mathbb{I}$ and $a_n = 0$. Let a_n be the smallest $n > 1$ such that $a_n = 0$. If $n = 2$, $b_2 = -\det(F) \neq 0$ so $f^2 = \iota$. Else, $b_n = -\det(F)a_{n-1} \neq 0$ so $f^n = \iota$. Hence, the order of $f \neq \iota$ is the smallest $n > 1$ such that $a_n = 0 \pmod{p}$.

Conversely, a recurrence equation can be understood in part mod p using a suitable Fractional Linear Transformation. Let $a_n = \alpha a_{n-1} + \beta a_{n-2}$ for $\beta \neq 0$. The matrix $M = \begin{pmatrix} \alpha & \beta \\ 1 & 0 \end{pmatrix}$ would have

the characteristic equation $\lambda^2 - \alpha\lambda - \beta = 0$. Hence, the first value for which $a_n = 0$ can be found by studying the function $f(x) = \frac{\alpha x + \beta}{x}$.

Additionally, the method for solving many of these recurrence relations can be understood by studying the matrix representation. Take a series where M is diagonalizable, that is, $M = PDP^{-1}$ for D being the diagonal matrix of the eigenvalues of D . Consider

$$\begin{pmatrix} \alpha & \beta \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix} = \begin{pmatrix} \alpha a_n + \beta a_{n-1} \\ a_n \end{pmatrix} = \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}$$

It follows that the values of a_n can be understood by $M^n v$ where v is the vector $[a_1, a_0]^T$. Now,

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a_1 \\ a_0 \end{pmatrix} = PD^n P^{-1}v$$

giving that $a_n = r\lambda_1^n + s\lambda_2^n$ where r, s are chosen so that $a_0 = r + s$ and $a_1 = r\lambda_1 + s\lambda_2$.

Two particularly nice examples of the use of the recurrence equation for understanding Fractional Linear Transformations are the functions $u(x) = \frac{x-1}{x+1}$ and $v(x) = \frac{x-3}{x+1}$. The associated characteristic polynomial for $u(x)$ is $\lambda^2 - 2\lambda + 2 = 0$, so the recurrence equation is $a_n = 2a_{n-1} - 2a_{n-2}$. For any p , the series proceeds as $a_n = 0, 1, 2, 2, 0, \dots$. Hence, for $p > 2$, a_n is zero first when $n = 4$ so $u(x)$ is composed of 4-cycles and potential fixed points.

Similarly, the associated characteristic polynomial for $v(x)$ is $\lambda^2 - 2\lambda + 4 = 0$, and the recurrence equation is $a_n = 2a_{n-1} - 4a_{n-2}$. For any p , the series proceeds as $a_n = 0, 1, 2, 0, \dots$. For $p > 2$, a_n is zero first when $n = 3$ so $u(x)$ is composed of 3-cycles and potential fixed points.

Solving $u(x) = x$ reduces to whether $x^2 = -1$ has a solution in $\mathbb{F}_p$, or in other words, whether -1 is a quadratic residue mod p . Since 4 is not prime, u has either 0 or 2 fixed points. Under the transformation of u , the $p+1$ elements of $\mathbb{P}_p$ split into 4-cycles and fixed points. If $p \equiv 1 \pmod{4}$, $\mathbb{P}_p$ does not divide evenly into 4-cycles so u has two fixed points and -1 is a quadratic residue mod p . If $p \equiv 3 \pmod{4}$, $\mathbb{P}_p$ splits evenly and -1 is not a quadratic residue mod p .

Similarly, solving $v(x) = x$ reduces to $x^2 = -3$ and tests if -3 is a quadratic residue. If $p \equiv 1 \pmod{3}$, $\mathbb{P}_p$ does not split evenly into 3-cycles and -3 is a quadratic residue mod p . If $p \equiv 2 \pmod{3}$, $\mathbb{P}_p$ splits evenly into 3-cycles and -3 is not a quadratic residue mod p .

Together, these lead us to the following theorem:

Theorem 5.1. *-1 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod{4}$ and -3 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod{3}$.*

A Group Theory

We begin by revisiting some properties of groups.

Take a subgroup of $H \leq G$. A left coset aH for some $a \in G$ is the set of elements $aH = \{ah | h \in H\}$. Since the a is arbitrary and the identity e of G is in H , it is easy to see that the cosets of H form a partition of G . Additionally, if H is not trivial, the cosets need not be distinct. The index of a subgroup $H \leq G$ is denoted by $[G : H]$ and is the number of distinct cosets of H in G .

A group action of G on a set S is a map $\phi : G \times S \rightarrow S$ that satisfies the following axioms:

- $\phi(e, s) = s$ for all elements $s \in S$ where e is the identity element of G
- $\phi(gh, s) = \phi(g, \phi(h, s))$ for all $g, h \in G$ and all elements $s \in S$

The first axiom states that there is an identity transformation and the latter states that the group action is compatible with group multiplication. Note that since $\phi(e, s) = \phi(g^{-1}, \phi(g, s)) = s$, the group action forms a bijection on S . Hence, the group action can be considered a homomorphism from G to the symmetric group of S . For ease of notation, we will use $g.s$ to denote $\phi(g, s)$.

A.1 Lagrange's Theorem

Theorem A.1. For a subgroup $H \leq G$, $[G : H]|H| = |G|$.

Proof. The left coset aH has the same number of elements as H . It is obvious that aH cannot have more elements than H . To see why it has as many elements as H , assume $ah_1 = ah_2$. Then, since $a \in G$, a has an inverse so it follows that $h_1 = h_2$ so multiplication by a forms a bijection on the sets H and aH .

Next, assume that $bh_1 \in aH$ for some $b \in G$ and $h_1 \in H$. It follows that for some $h_2 \in H$, $bh_1 = ah_2$ or in other words, $b^{-1}a = h_1h_2^{-1} = h' \in H$. Now, consider an element bh in the coset bH :

$$bh = b(h')^{-1}h'h = bb^{-1}ah'h = ah'h \in aH$$

thus showing cosets are disjoint subsets of G . The theorem follows because the cosets partition G into disjoint cosets, each of size $|H|$. $\square$

A.2 Orbit-Stabilizer Theorem

For a group action of G on a set S , define the stabilizer of an element $s \in S$ as the set $\text{Stab}(s) = \{g \in G \mid g.s = s\}$ and the orbit of an element $s \in S$ as the set $\text{Orb}(s) = \{s' \in S \mid \exists g \in G, g.s = s'\}$. Hence, the stabilizer of s is all the group elements that have s as a fixed point and the orbit of s is the set of all possible images of s by transformations of the group action.

Theorem A.2. For a group action G on a set S and any element $s \in S$, $|\text{Stab}(s)| \cdot |\text{Orb}(s)| = |G|$.

Proof. We begin by showing $\text{Stab}(s)$ is a subgroup of G . Clearly, the identity element fixes s so that $e.s = s$. If g and h fix s , $gh.s = g.h.s = g.s = s$ so gh also fixes s . Lastly, $s = e.s = g^{-1}.g.s = g^{-1}.s$ so g^{-1} fixes s .

We know from Lagrange's theorem that $|G| = [G : \text{Stab}(s)]|\text{Stab}(s)|$. We now show a bijection between $\text{Orb}(s)$ and the left cosets of $\text{Stab}(s)$. Consider two elements $g, g' \in G$ such that $g.s = g'.s = a \in \text{Orb}(s)$. We have that this is possible if and only if $g^{-1}g'.s = s$ or in other words, $g^{-1}g' \in \text{Stab}(s)$. This immediately implies that the coset $g\text{Stab}(s)$ is the same as $g'\text{Stab}(s)$. Conversely, assume g, g' are in the same coset of $\text{Stab}(s)$. This implies that $g^{-1}g' \in \text{Stab}(s)$ from which it follows that $g.s = g'.s$. This establishes the necessary bijection and completes the proof. $\square$

B Linear Algebra

B.1 The order of the General Linear Group

Theorem B.1. The order of the group $\text{GL}_2(\mathbb{F}_p)$ is $(p-1)^2p(p+1)$

Proof. Consider a 2×2 invertible matrix over $\mathbb{F}_p$ as the collection of two linearly independent vectors:

- For the first vector, we have p choices for each of the two entries of the vector. However, we have the restriction that not both the entries are zero. Hence, this gives $p^2 - 1$ possibilities.
- For the second vector, we once more have p choices for each of the two entries of the vector. Since the second vector is linearly independent to the first, it cannot be the first vector scaled by any element of $\mathbb{F}_p$. Removing the p vectors obtained by scaling the first vector, we get $p^2 - p$ possibilities.

Thus, we have:

$$|\text{GL}_2(\mathbb{F}_p)| = (p^2 - 1)(p^2 - p) = (p-1)^2p(p+1)$$

which completes the proof. $\square$

B.2 The Characteristic Equation

Theorem B.2. For an eigenvector v of a matrix M with corresponding eigenvalue $\lambda \in K$, λ is a root of the characteristic polynomial $\det(M - \lambda\mathbb{I}) = 0$.

Proof. Since v is an eigenvector of M , $Mv = \lambda v = \lambda(\mathbb{I}v)$ so that $(M - \lambda\mathbb{I})v$ is the zero vector. Since the null space of $(M - \lambda\mathbb{I})$ is non-trivial (it is at least the span of v), $\det(M - \lambda\mathbb{I}) = 0$. It also follows that if M is a degree n matrix, $\det(M - \lambda\mathbb{I})$ is a degree n polynomial over K so there are at most n eigenvalues. $\square$

Theorem B.3. If M is a matrix over K and $\lambda \in K$ solves the characteristic equation of M , then M has at least one eigenvector with corresponding eigenvalue λ .

Proof. Since λ solves the characteristic equation of M , $\det(M - \lambda\mathbb{I}) = 0$ and the null space of $M - \lambda\mathbb{I}$ is non-trivial. Any vector v in this null space solves $Mv = \lambda v$ and so v is an eigenvector of M with eigenvalue λ . $\square$

B.3 Independence of Eigenvectors

Theorem B.4. If λ_1, λ_2 are distinct eigenvalues of degree 2 matrix M over K , the corresponding eigenvectors, v_1, v_2 , are linearly independent and form a basis of K^2 .

Proof. Assume there is some nontrivial linear relation between v_1, v_2 i.e. there exist non-zero a, b so that $av_1 + bv_2 = 0$. Act on both sides of the equation with $M - \lambda_1\mathbb{I}$. Then,

$$(M - \lambda_1\mathbb{I})(av_1 + bv_2) = 0$$

$$b(\lambda_2 - \lambda_1)v_2 = 0$$

since v_1 is in the null space of $M - \lambda_1\mathbb{I}$. Since v_2 is non-zero and $\lambda_2 \neq \lambda_1$ by assumption, b must be 0. But then it follows that a must also be 0 so v_1, v_2 are linearly independent. Hence, v_1, v_2 form a two dimensional subspace of K^2 and must be a basis of K^2 . $\square$

Theorem B.5. If degree 2 matrix M over K has only one eigenvalue λ (of multiplicity 2), it is either a constant multiple of the identity matrix or else has only one eigenvector (up to scaling).

Proof. Since M has one eigenvalue, it has at least one eigenvector with eigenvalue λ . If M has two linearly independent eigenvectors v_1, v_2 , then v_1, v_2 forms a basis of K^2 . Let $v = av_1 + bv_2$ be an arbitrary vector in K^2 . Then, $Mv = M(av_1 + bv_2) = aMv_1 + bMv_2 = a\lambda v_1 + b\lambda v_2 = \lambda v$. Hence, M sends any vector v to λv i.e. $M = \lambda\mathbb{I}$. $\square$

B.4 Diagonalization

Theorem B.6. If M is a degree 2 matrix over K and has two distinct eigenvalues λ_1, λ_2 with corresponding eigenvectors v_1, v_2 , it can be decomposed into the form PDP^{-1} where D is a diagonal matrix with entries λ_1, λ_2

Proof. Since the eigenvalues are distinct, v_1, v_2 form a basis of K^2 and any linear transformation is determined uniquely by its action on v_1, v_2 . Take P to be the matrix with columns v_1, v_2 . Then, P takes $[1, 0]^T$ to v_1 and $[0, 1]^T$ to v_2 . It is easy to verify then that if D is the diagonal matrix with entries λ_1, λ_2 , $(PDP^{-1})v_1 = \lambda_1 v_1$ and $(PDP^{-1})v_2 = \lambda_2 v_2$. Then, M and PDP^{-1} have the same action on v_1, v_2 and hence must be equal. $\square$

B.5 Cayley-Hamilton Theorem

Theorem B.7. Any degree 2 invertible matrix M satisfies its own characteristic equation i.e. $M^2 - \text{tr}(M)M + \det(M)\mathbb{I} = 0$. This is a special case of the general Cayley-Hamilton Theorem which states that any matrix satisfies its own characteristic equation, but it suffices for the purposes needed.

It will first be useful to outline some properties of conjugation by a matrix.

Lemma B.8. Conjugation preserves addition and multiplication of matrices.

Proof. Let A, B, P be invertible matrices. It is easy to show that $PAP^{-1} + PBP^{-1} = P(A + B)P^{-1}$ and $PAP^{-1}PBP^{-1} = PABP^{-1}$. $\square$

Lemma B.9. Conjugation preserves the trace and the determinant of a matrix.

Proof. Let A be a matrix with entries A_{ij} and B be a matrix with entries B_{ij} . The trace of A is the sum of the elements of the diagonal i.e. $\text{tr}(A) = \sum_i A_{ii}$. The matrix AB has entries $(AB)_{ik} = \sum_j A_{ij}B_{jk}$ while the matrix BA has entries $(BA)_{ik} = \sum_j B_{ij}A_{jk}$. Hence,

$$\text{tr}(AB) = \sum_i \sum_j A_{ij}B_{ji} = \sum_j \sum_i B_{ji}A_{ij} = \text{tr}(BA)$$

Now, let P be an invertible matrix. $\text{tr}((PA)P^{-1}) = \text{tr}(P^{-1}(PA)) = \text{tr}(A)$ thus showing that conjugation preserves the trace.

To show that conjugation by a matrix preserves the determinant, we use the multiplicative property of matrices i.e. $\det(AB) = \det(A)\det(B)$. The multiplicative property can be understood by viewing $\det(A)$ as the factor that scales the volume of the parallelotope formed by some set of basis vectors. Then AB has the effect of first scaling the volume by B then by A . Now, let P be an invertible matrix.

$$\begin{aligned} \det(PAP^{-1}) &= \det(P)\det(A)\det(P^{-1}) \\ &= \det(A)\det(P)\det(P^{-1}) \\ &= \det(A)\det(PP^{-1}) = \det(A) \end{aligned}$$

$\square$

Now we are ready to prove the theorem.

Proof. For a matrix $M \in \text{GL}_2(\mathbb{F}_p)$, our goal will be to choose a sufficiently nice conjugated matrix PMP^{-1} that satisfies its characteristic polynomial. Consider the characteristic polynomial of PMP^{-1} :

$$(PMP^{-1})^2 - \text{tr}(PMP^{-1})PMP^{-1} + \det(PMP^{-1})\mathbb{I} = 0$$

$$PM^2P^{-1} - \text{tr}(M)PMP^{-1} + \det(M)\mathbb{I} = 0$$

$$P(M^2 - \text{tr}(M)M + \det(M)\mathbb{I})P^{-1} = 0$$

$$M^2 - \text{tr}(M)M + \det(M)\mathbb{I} = P^{-1}(0)P = 0$$

which implies that M satisfies its characteristic equation if and only if PMP^{-1} satisfies its characteristic equation.

Assume that $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ for $c \neq 0$. Then, let $P = \begin{pmatrix} c & -a \\ 0 & 1 \end{pmatrix}$. Direct computation gives $PMP^{-1} = \begin{pmatrix} 0 & -\det(M) \\ 1 & \text{tr}(M) \end{pmatrix}$. For simplicity of notation, let $u = -\det(M)$, $v = \text{tr}(M)$ and $Q = PMP^{-1}$ so that $Q = \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix}$.

We show that $Q^2 - \text{tr}(Q)Q + \det(Q)\mathbb{I}$ sends both $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -v \\ 1 \end{pmatrix}$ to the zero vector. Since these vectors form a basis, the transformation must send every vector to zero and hence, be the zero matrix.

$$Q : \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} u \\ v \end{pmatrix}$$

$$Q : \begin{pmatrix} -v \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} u \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ u \end{pmatrix}$$

Hence,

$$Q^2 - \text{tr}(Q)Q + \det(Q)\mathbb{I} : \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} u \\ v \end{pmatrix} - v \begin{pmatrix} 0 \\ 1 \end{pmatrix} - u \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$Q^2 - \text{tr}(Q)Q + \det(Q)\mathbb{I} : \begin{pmatrix} -v \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ u \end{pmatrix} - v \begin{pmatrix} u \\ 0 \end{pmatrix} - u \begin{pmatrix} -v \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Next, assume $M = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ for $b \neq 0$. Let $P = \begin{pmatrix} ad & 0 \\ -a & -b \end{pmatrix}$. Direct computation gives $Q = PMP^{-1} = \begin{pmatrix} 0 & -ad \\ 1 & a+d \end{pmatrix} = \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix}$. Identical computation to the first case gives that matrices M of this form also satisfy their characteristic equation.

Lastly, assume $M = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$. Direct computation of $M^2 - \text{tr}(M)M + \det(M)\mathbb{I}$ gives:

$$\begin{aligned} M^2 - \text{tr}(M)M + \det(M)\mathbb{I} &= \begin{pmatrix} a^2 & 0 \\ 0 & d^2 \end{pmatrix} - (a+d) \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} + ad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} a^2 - a(a+d) + ad & 0 \\ 0 & d^2 - d(a+d) + ad \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

□

We now consider some nice generalizations of the methods presented in this proof. The method for proving the Cayley-Hamilton theorem for matrices of the form $\begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix}$ by showing that they map a basis to the zero vector can be extended to some higher dimensional matrices. Let e_i be the standard orthonormal basis for F^n . Let M be the matrix with columns $e_2, e_3, \dots, e_{n-1}, v$ with $v = [v_1, \dots, v_n]^T$. Then, M has the characteristic polynomial $\lambda^n - v_n\lambda^{n-1} - \dots - v_2\lambda - v_1$. Choose the basis $B = \{e_1, -v_n e_1 + e_2, -v_{n-1} e_1 - v_n e_2 + e_3, \dots\}$. Then, it can be shown that if $f(\lambda) = \det(M - \lambda\mathbb{I})$, $f(M)$ sends all vectors in B to 0.

Let us look at a 4×4 matrix as an example.

$$M = \begin{pmatrix} 0 & 0 & 0 & d \\ 1 & 0 & 0 & c \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & a \end{pmatrix}$$

consider the basis

$$B = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -a \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -b \\ -a \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -c \\ -b \\ -a \\ 1 \end{pmatrix} \right\}$$

Then,

$$\begin{aligned}
M : \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} &\rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} d \\ c \\ b \\ a \end{pmatrix} \\
M : \begin{pmatrix} -a \\ 1 \\ 0 \\ 0 \end{pmatrix} &\rightarrow \begin{pmatrix} 0 \\ -a \\ 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ -a \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} d \\ c \\ b \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ d \\ c \\ b \end{pmatrix} \\
M : \begin{pmatrix} -b \\ -a \\ 1 \\ 0 \end{pmatrix} &\rightarrow \begin{pmatrix} 0 \\ -b \\ -a \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} d \\ c \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ d \\ c \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ d \\ c \end{pmatrix} \\
M : \begin{pmatrix} -c \\ -b \\ -a \\ 1 \end{pmatrix} &\rightarrow \begin{pmatrix} d \\ 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ d \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ d \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \\ d \end{pmatrix}
\end{aligned}$$

The characteristic polynomial (in terms of M) ends up being $M^4 - aM^3 - bM^2 - cM - d\mathbb{I}$. It is easy to verify that the transformation of the characteristic polynomial sends all vectors in B to 0.